

Introductory Concepts in Nutrition

Topic 6

Nutritional assessment: Direct and indirect methods

Susmita Ghosh, MS

Dr. Ratthiphone Oula, MD, MCN

Dr. Nilupa S. Gunaratna, MS, PhD

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Learning objectives

By the end of topic 6, participants will be able:

1. To explain the importance of nutritional assessment for indicating nutritional well-being.
2. To recognize different nutritional assessment methods and their purposes and applications.
3. To explain how nutritional assessment helps to indicate the nutritional status of an individual, community, or population group, track healthy growth of children, and identify risk of non-communicable diseases.

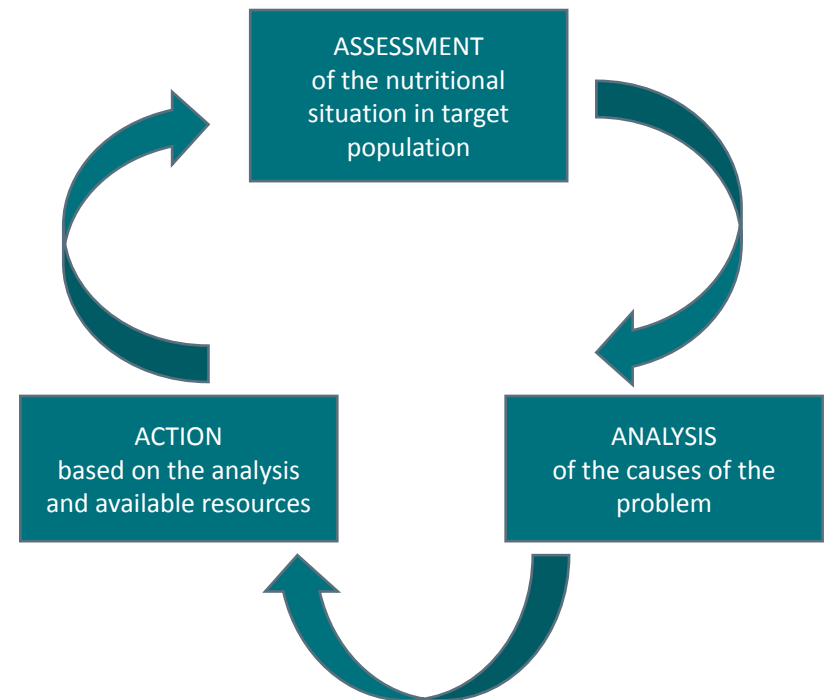
Nutritional status

- Nutritional status is the current body status of a person or a population group, related to their state of nourishment (the consumption and utilization of nutrients)
- Nutritional status is determined by a complex interaction between internal (constitutional) factors and external (environmental) factors:
 - Internal factors include age, sex, eating behavior, physical activity, and diseases
 - External factors include food safety, culture, and social and economic circumstances

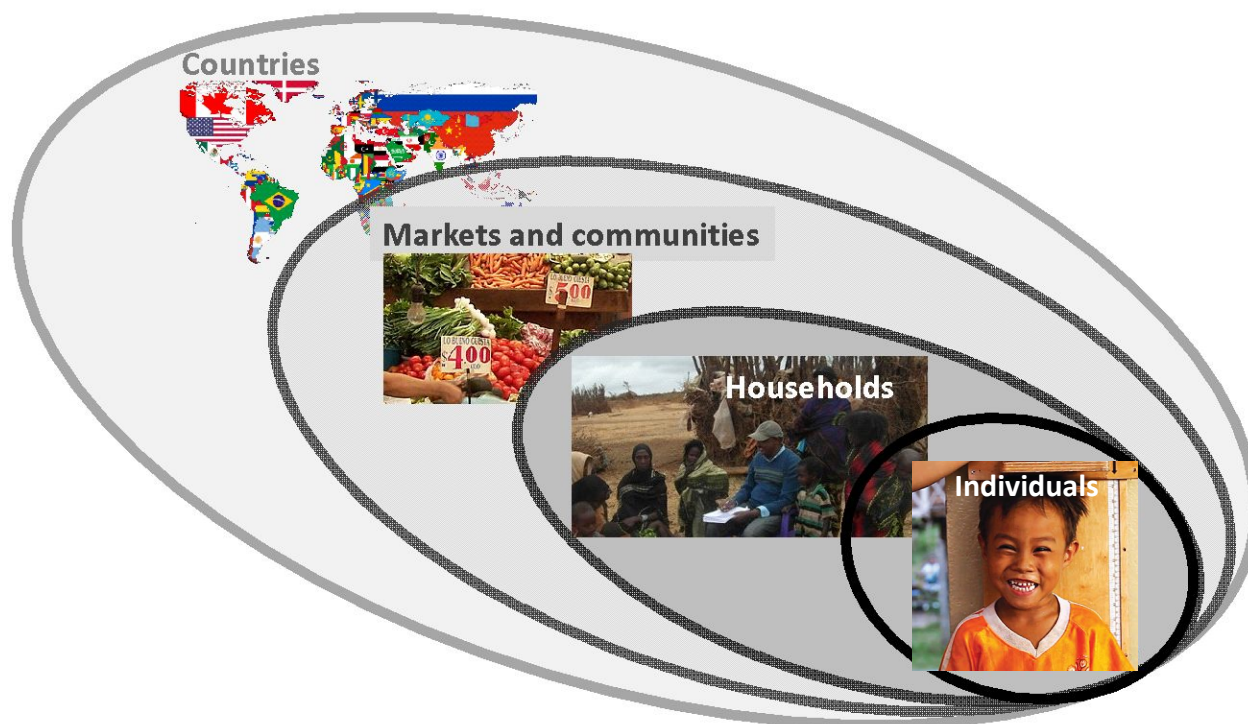
Nutritional assessment

- **Nutritional assessment** is the interpretation of anthropometric, biochemical (laboratory), clinical, and dietary data to determine whether a person or group of people are well-nourished or malnourished
- Nutritional status is assessed by direct and indirect methods
 - Direct methods deal with individuals, while indirect methods use community indices that reflect the community's nutritional status

Nutritional assessment and analysis



Nutritional assessment at different levels



Data can be available at multiple levels: countries, communities, households, and individuals. Some examples:

- Countries: Food balance sheets are constructed using country-level data for individual food commodities
- Communities: Food prices are observed at markets where products are bought and sold
- Households: Household food consumption
- Individuals: Anthropometric (body) measurements or biochemical measurements (biomarkers)

Purpose and application of nutritional assessment

Nutritional assessment is used:

- To identify individuals or population groups at risk of becoming malnourished
- To develop health care programs that address community needs as identified by the evaluation
- To assess the efficacy of the nutritional interventions and programs

Different types of direct nutritional assessment

Nutritional assessment	Example of nutritional indicator	Application
Anthropometry	Height/length for age, weight for height, weight for age, stunting, wasting, & underweight	Assess child growth and development
Biochemical/ laboratory	Prevalence of serum/plasma retinol <0.70 micromol/L in pre-school age children ○ <2% ○ 2-9% ○ 10-19% ○ ≥20%	Determine level of public health problem: ○ None ○ Mild ○ Moderate ○ Severe
Clinical	Proportion of pregnant women with nightblindness ≥ 5%	Identify moderate public health problem
Dietary	Proportion of children consuming an adequate amount of vitamin A	Quantify dietary quality and nutrient adequacy

Anthropometric assessment

- Provides a valuable assessment of nutritional status in a population (children or adults).
- Rationale for children: All of the world's children grow similarly, if they are fed well, receive good care, and remain free of infectious diseases and deprivations
- Anthropometric measurements are quantitative, noninvasive, less expensive, and require less training, equipment, and supplies
- The core elements of anthropometry are height (or recumbent length for very young children), weight, and body proportions.
- Different anthropometric measures such as waist circumference (WC), hip circumference (HC), body mass index (BMI), waist-to-hip ratio, and waist-to-height ratio are commonly used to predict risk of non-communicable disease such as diabetes and hypertension.

Anthropometric measurement

- **Weight:** Weight is measured for children and as well as for adults to identify the normal growth for children and find out the risk of non-communicable disease for adults.
 - Calibrated beam or a digital infant scale is used to measure weight for children
 - A balanced floor scale or electronic floor scale is used to measure weight for adults
- **Recumbent length:** Recumbent length is measured for children under 2 years of age.
- **Height:** Height is measured for children 2 years and older who are more than 87 cm tall and for adults.
 - Measuring height requires a standard measuring board with a moveable headboard (stadiometer) attached to a flat wall. Length is measured on a flat surface with a movable footboard.

Anthropometric measurement

- **Head circumference:** Head circumference is a measurement of a child's head around its largest area
 - The head circumference of infants increases rapidly in the first 2 years of life
 - Increase in head circumference in the first 2 years of life is affected by nutritional status and non-nutritional factors
- **Mid upper arm circumference:** MUAC is a measure of the sum of the muscle and subcutaneous fat in the upper arm.
 - With acute malnutrition, both fat and muscle are reduced in the upper arm
 - It can be used as an indicator for body composition in adults and children

Anthropometric measurement

- **Hip and waist measurement:** Waist circumference is measured halfway between the lower ribs and the iliac crest, while hip circumference is measured at the largest circumference around the buttocks.
 - Waist circumference is related to insulin resistance
 - A waist circumference greater than 94 cm in men and 80 cm in women has been associated with increased risk for cardiovascular disease

Nutritional indices

- Most single anthropometric measurements do not specifically assess nutritional status
- Nutritional indices are derived by combining two or more anthropometric measurements
- Different types of nutritional indices are:
 - Body mass index (BMI)
 - Waist-to-hip ratio (WHR)
 - Weight-for-height z-score

Nutritional indices

- **Body mass index (BMI):** an anthropometric indicator based on weight-to-height ratio. It is estimated by dividing weight (kg) by height squared (m^2).
 - Adults are considered malnourished if their BMI is less than 18.5, normal from 18.5 to 24.9, and overweight if ≥ 25
- **Weight-for-height:** an indicator of body composition.
 - Depends on sex and age
- **Waist-to-hip ratio (WHR):** Waist circumference divided by hip circumference.
 - Cutoffs for WHR are ≥ 0.85 for women and ≥ 1.00 for men

Biochemical methods

- Biochemical data can identify or diagnose a nutritional problem, including nutrient deficiencies and/or toxicities, in an individual.
- It can also be used to assess the nutritional status of a population.
- Biochemical measurements can reflect the immediate past intake of nutrients, or the changes produced by a long-standing deficient intake.
- Biochemical parameters can be used for acute or subclinical conditions when other methods of nutritional assessment fail to identify a condition.
- Some biochemical measurements are affected by other factors (for example, inflammation)

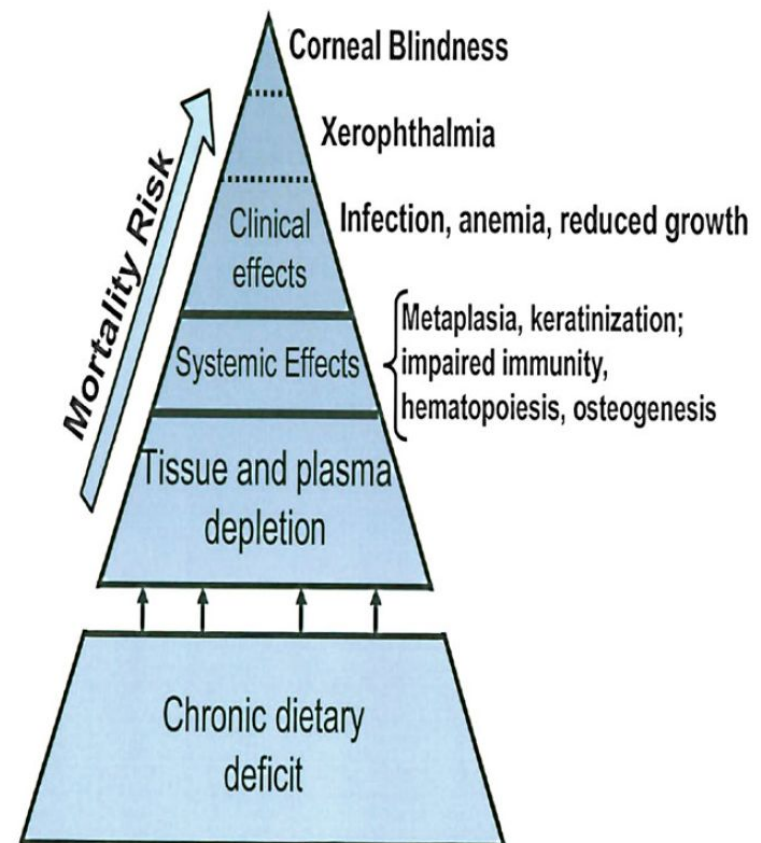
Biochemical methods

- Biochemical assessment includes two types of test: static and functional.
- Static tests include:
 - Measurement of nutrient concentration in blood, urinary excretion and urinary metabolites of a nutrient, abnormal metabolites in urine/blood due to nutrient deficiency
 - Measurement of changes in blood constituents or enzyme activities
 - Measurement of 'tissue specific' chemical markers
 - Saturation, loading, and isotope tests
- Functional tests reflect the failure of function or physiologic process as a result of nutritional deficiency.
 - For example: visual adaption to darkness will be compromised by vitamin A deficiency

Stages of vitamin A deficiency

- Mild to moderate deficiencies can have no clinical signs but still affect a person's well-being and development.
- Clinical signs appear only with extreme micronutrient deficiency.
- Clinical signs observed in a population indicate that a substantial proportion of the population suffers from deficiency, affecting health and development.

Vitamin A Deficiency



Evaluation of vitamin A status

- Vitamin A deficiency results from a dietary intake of vitamin A that is inadequate to satisfy physiological needs.
- It may be exacerbated by high rates of infection, especially diarrhea and measles
 - Retinol is the main circulating form of vitamin A in blood and plasma
 - Serum retinol levels reflect liver vitamin A stores when they are severely depleted or extremely high
 - Serum retinol is best used for the assessment of subclinical/biochemical vitamin A deficiency in a population (not in an individual)
 - A plasma or serum retinol concentration $<0.70 \mu\text{mol/L}$ indicates subclinical vitamin A deficiency in children and adults, and a concentration of $<0.35 \mu\text{mol/L}$ indicates severe vitamin A deficiency

Evaluation of iodine status

- Urinary iodine concentration is currently the most practical biochemical marker for iodine status.
- This approach assesses iodine status only at the time of measurement, whereas thyroid size reflects iodine nutrition over months or years.
- Therefore, even though populations may have attained iodine sufficiency on the basis of median urinary iodine concentration, goiter may persist, even in children.

Epidemiological criteria for assessing iodine nutrition based on median urinary iodine concentrations of school-age children (≥ 6 years)

Median urinary iodine ($\mu\text{g/l}$)	Iodine intake	Iodine status
< 20	Insufficient	Severe iodine deficiency
20–49	Insufficient	Moderate iodine deficiency
50–99	Insufficient	Mild iodine deficiency
100–199	Adequate	Adequate iodine nutrition
200–299	Above requirements	Likely to provide adequate intake for pregnant/lactating women, but may pose a slight risk of more than adequate intake in the overall population
≥ 300	Excessive	Risk of adverse health consequences (iodine-induced hyperthyroidism, autoimmune thyroid diseases)

*Applies to adults, but not to pregnant and lactating mother

Evaluation of iron status

- Heme iron may be found in meat, poultry, fish
- Non-heme iron may be found in cereal, greens, peas/beans, eggs, dried fruits
- Vitamin C improves iron absorption. Vitamin C may be found in fruits such as oranges, guava, lemon, etc.
- Stages of iron deficiency:
 - Depletion present with decreased **serum ferritin**
 - Iron deficiency present with decreased **transferrin** saturation
 - Iron deficiency anemia present with decreased **hemoglobin** (Hgb) and decreased **mean corpuscular volume (MCV)**

Indicators used to detect iron deficiency and iron deficiency anemia

Assessment of iron deficiency and iron deficiency anemia

Indicator	Measure	Cutoffs	Indication	Commonly used methods	Special considerations
Serum ferritin	Total body iron stores	<15 µg/L <12 µg/L	Depleted iron stores in individuals 5 years or older Depleted iron stores in individuals less than 5 years of age	Venous or capillary blood, dried blood spots (DBS) ELISA method	Infection and inflammation may cause inflated ferritin values Use of DBS convenient for field work
Serum transferrin concentration (TIBC)	Concentration of iron transport protein	360 µg/dL	Depleted iron stores	Venous blood Assay in which transferrin is saturated with excess iron; chromogenic methods	Influenced by infection and inflammation More complicated laboratory procedure that requires quality-control sera
Serum transferrin saturation	Iron transport protein	<15%	Iron-deficient erythropoiesis	Venous blood Calculated from TIBC and serum iron values	Influenced by infection and inflammation Diurnal variation
Soluble Serum transferrin receptor (STfR)	Expression of STfR, which bind ferritin for uptake in cells	High (10 mg/L)	Iron-deficient erythropoiesis	Venous blood ELISA method	Possible to quantify STfR using a dried blood spot Not significantly influenced by infection and inflammation Can be influenced by other nutritional deficiencies such as B12 and folate deficiency and specifically by acute malaria infection
Free erythrocyte protoporphyrin (FEP)	Serves as a precursor in heme biosynthesis	>70 mmol FEP/mol heme	Iron-deficient erythropoiesis	Whole blood (drop) Hematofluorometry	Influenced by infection and inflammation Portable hematofluorometer available
Hemoglobin (SCN 2000)	Blood hemoglobin concentration	<11.0 g/dL <12.0 g/dL	Anemia in pregnant women Anemia in nonpregnant women >15 year	Venous or capillary blood Dried blood spot HemoCue or cyanmethemoglobin method	Influenced by certain parasitic infections and other micronutrient deficiencies It is necessary to make adjustments to cutoff values for persons living in high altitudes Less expensive, field friendly equipment available
Hematocrit	Packed red blood cell volume	<0.36 <0.33	Anemia in nonpregnant women >15 year Anemia in pregnant women	Whole blood Centrifugation method	Less expensive, but methods can be difficult to standardize in a field setting
Erythrocyte	Color and shape or erythrocyte	Microcytic or hypochromic	Anemia	Whole blood Microscopy	

Evaluation of iron status

- **Serum ferritin** combines with Fe and is stored in liver, spleen, and bone marrow.
 - Most sensitive and best test to detect early iron deficiency
- **Transferrin** carries Fe in blood. Associated and used to calculate TIBC (total iron binding capacity).
 - When TIBC goes up, iron level is low. When TIBC goes down, iron level is high.
- **Hemoglobin (Hgb)** is the molecule in red blood cells (RBCs) that holds iron and allows the cells to carry oxygen to the body tissues.
- **Hematocrit (Hct)** is expressed as RBCs as a percentage of total blood volume.

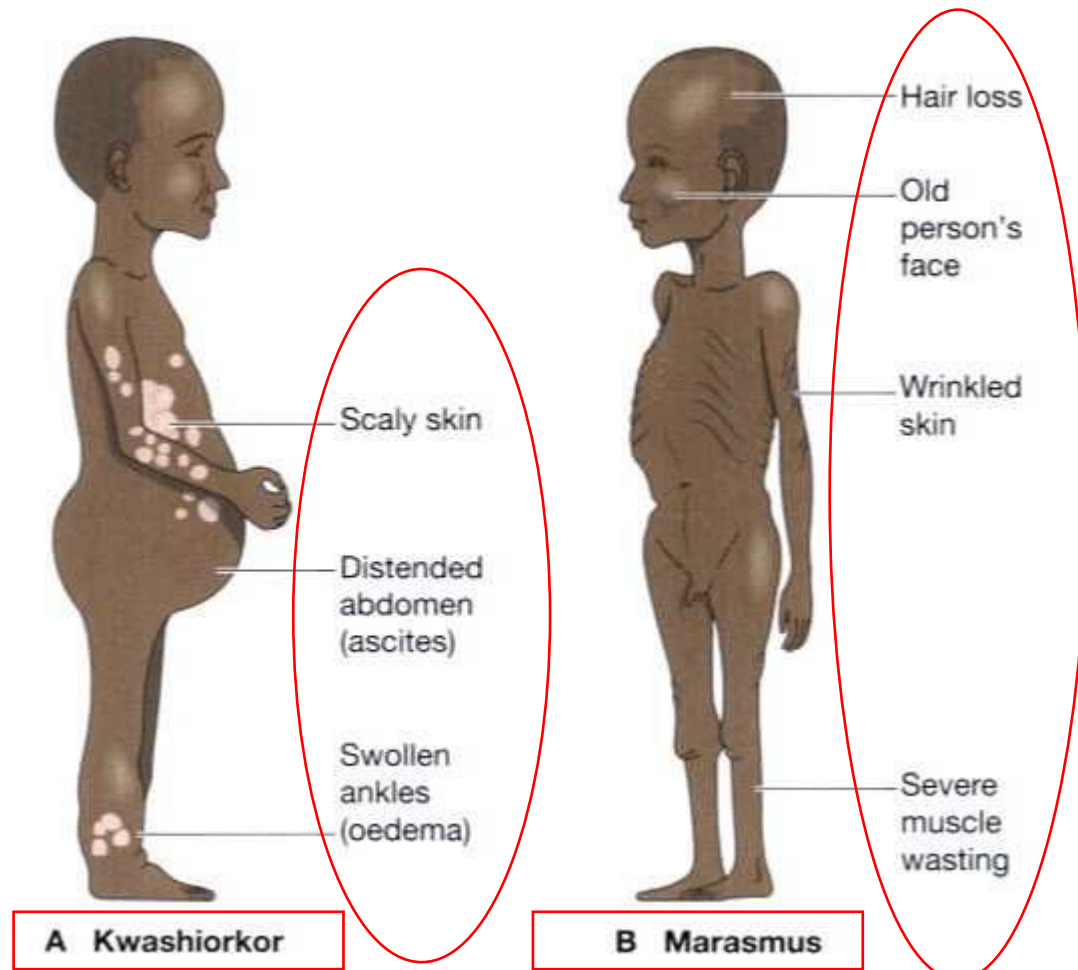
Limitations of biochemical methods

- Measurement error is possible; measurement error is the difference between the true value of the biomarker and the measured value.
- Measurement error can be happened either by biological (preanalytical) or laboratory (analytical) error.
- Some biomarkers can have day-to-day or even within-day variability.
- Some nutrients do not have good, specific biomarkers.
- Biochemical assessments require training, equipment, and supplies, which can be costly.
- As with all nutritional assessment, consent is important, and biochemical assessments in particular can be invasive.

Clinical methods

- Clinical assessment includes checking for visible signs of nutritional deficiencies such as bilateral pitting edema, emaciation (a sign of wasting, which is loss of muscle and fat tissue as a result of low energy intake and/or nutrient loss from infection), hair loss, and changes in hair color.
- Clinical nutrition assessment also includes checking for or asking clients about symptoms of infection that can increase nutrient needs (e.g., fever) and nutrient loss (e.g., diarrhea and vomiting), as well as medical conditions (e.g., HIV, celiac disease) that impair digestion and nutrient absorption and increase the risk of developing malnutrition

Sign and symptoms of marasmus and kwashiorkor



Clinical signs and symptoms of micronutrient deficiency

Selected nutrients	Deficiency signs and symptoms
Vitamin A (retinol)	• Night blindness • Various stages of xerophthalmia • Hair follicle blockage with a permanent “goose-bump” appearance, follicular hyperkeratosis • Dry, rough skin
Vitamin D (calciferol)	• Rickets • Osteomalacia and osteoporosis • Epiphyseal swelling
Vitamin K (phylloquinone)	• Small hemorrhages in the skin or mucous membranes (petechia) • Intraocular hemorrhage
Vitamin B1 (thiamin)	• Beriberi (cardiac and neurologic) • Wernicke and Korsakov syndromes (alcoholic confusion and paralysis) • Sensory loss, blurred vision • Dyspnea • Muscular wasting • Sometimes edema (wet beriberi) • Malaise • Ataxia • Reduced tendon jerks (reflex) • Mental confusion • Tense calf muscles • Distended neck veins • Jerky movement of eyes • Staggering gait and difficulty walking • Infants may develop cyanosis • Round, swollen (moon) face • Foot and wrist drop

Clinical signs and symptoms of micronutrient deficiency (continued)

Vitamin B2 (riboflavin)	• Redness and scaling of nasolabial folds • Diffuse depigmentation • Non specific—fatigue, eye changes, dermatitis, brain dysfunction, impaired iron absorption • Tearing, burning, and itching of the eyes • Fissuring in the corners of the eyes • Soreness and burning of the lips, mouth, and tongue with fissuring and/or cracking of the lips and corners of the mouth • Purple swollen tongue • Seborrhea of the skin in the nasolabial folds, scrotum, or vulva • Capillary overgrowth around the corneas
Vitamin B3 (niacin)	• Pellagra identified by the four Ds: dermatitis, dementia, diarrhea and death if not treated. The dermatitis from niacin deficiency has classical presentation of dermatitis that is symmetrical and occurs at pressure points (e.g., buttocks) and on sun exposed skin • Sensory loss • Tremors • Sore tongue • Amblyopia • Anorexia • Indigestion
Vitamin B6 (pyridoxine)	• Dermatitis • Neurological disorders, convulsions • Anemia • Inflammation of the lining of the mouth, tongue inflammation • Fissures in the corners of the mouth

Clinical signs and symptoms of micronutrient deficiency (continued)

Selected nutrients	Deficiency signs and symptoms
Vitamin B9 (folic Acid)	• Glossitis • Neural tube defects • Weakness, fatigue, and depression • Pallor • Dermatologic lesions
Vitamin B12 (cobalamin)	• Lemon-yellow tint to the skin and eyes • Smooth, red, thickened tongue • Pallor • Ataxia
Vitamin C (ascorbic acid)	• Scurvy (fatigue, hemorrhages, low resistance to infection, anemia) • Edema • Swollen, bleeding, and/or retracted gums or tooth loss; mottled teeth; enamel erosion • Painful subperiosteal haematoma • Lethargy and fatigue • Skin lesions • Small red or purplish pinpoint discolorations on the skin or mucous membranes (petechiae) • Intraocular hemorrhage • Darkened skin around the hair follicles, • Corkscrew hair or spiral and unemerged hairs
Calcium	• Decreased bone mineralization, rickets, osteoporosis
Chromium	• Corneal lesions

Clinical signs and symptoms of micronutrient deficiency (continued)

Copper	• Hair and skin depigmentation • Pallor
Fluoride	• Increased dental decay, affects on bone health
Iodine	• Goiter • Developmental delay • Mental retardation • Hypothyroidism
Iron	• Fatigue • Decreased cognitive function • Headaches • Glossitis • Nail changes, koilonychias, thin, concave nails with raised edges • Skin pallor • Pale conjunctiva • Fatigue
Magnesium	• Tremors, muscle spasms, and tetany • Personality changes
Selenium	• Cutaneous alterations can occur including xerosis, erythematous scaly papules
Zinc	• Dwarfism and hypogonadism • Hepatosplenomegaly • Hyperpigmentation • Acrodermatitis enteropathica • Alopecia (hair loss) • Acral rash • Skin and eye lesions • Nasolabial seborrhea • Decubitus ulcers

Adapted from: Overview_of_micronutrient_deficiency_disorders_and_clinical_signs.ppt. www.micronutrient.org/nutritiontoolkit/ModuleFolders/3.Indicators%5CClinical%5CTools%5C

Gallagher [136]

Marks [137]

Allen et al. [138]

Tulchinsky [139]

Dietary methods

- Dietary assessment provides information on dietary quality and quantity of macro- and micronutrients consumed by an individual or population.
- Dietary assessment can be used to determine excess intakes (especially of calories) as well as nutrient deficiencies from a poor diet.
- Dietary assessment can identify risk factors that can lead to non-communicable diseases such as diabetes, cardiovascular diseases, and cancer.
 - Salt, sugar, fat, ultra-processed foods

Dietary assessment methods

- Dietary assessment methods
 - Direct
 - Retrospective
 - 1. Dietary
 - 24-hour recall
 - 2. Food frequency questionnaire
 - 3. Indicators of dietary quality
 - Innovative technologies
 - 1. Mobile-based technologies
 - 2. Automated multiple pass method (AMPM)
 - 3. Image-assisted dietary assessment methods
 - Prospective
 - 1. Estimated food record
 - 2. Weighed food record
 - Indirect
 - National
 - Food balance sheets
 - Household
 - Household consumption and expenditure surveys

24-hour recall

- A 24-hour dietary recall (24HR) is a structured interview intended to capture detailed information about all foods and beverages eaten by a respondent in the past 24 hours.
- This can be done via a face-to-face interview or telephone.
- The interview is often structured, with a standardized and pretested protocol involving probing questions that help the respondent remember all the items consumed and relevant food and beverage descriptions.
- Information on food type, preparation methods, recipes, brand names of commercial products, and estimated portion size are usually required.
- Trained staff must conduct the interview to prompt for details, such as cooking methods, portion sizes, beverages, snacking, and consumption outside the home.

Portion size used in dietary assessment

- Generally, individuals are unable to describe portion size accurately
- They would have difficulties conceptualizing serving sizes, thus standardized portion sizes are used
- To help quantify the amount of a food consumed, use:
 - Portion size estimation aids (photographs, everyday reference objects, household utensils, 3-D food models, containers with rice or beans)
 - Categorical sizes (small, medium, large)
 - Measurement using household utensils and food units (1 slice, 1 egg)
 - Standard units of measurement (gram, ounce, millimeters)
 - Other quantifying components (e.g., food models, cardboard cut-outs, etc.)

Portion size and toolkits used in dietary assessment

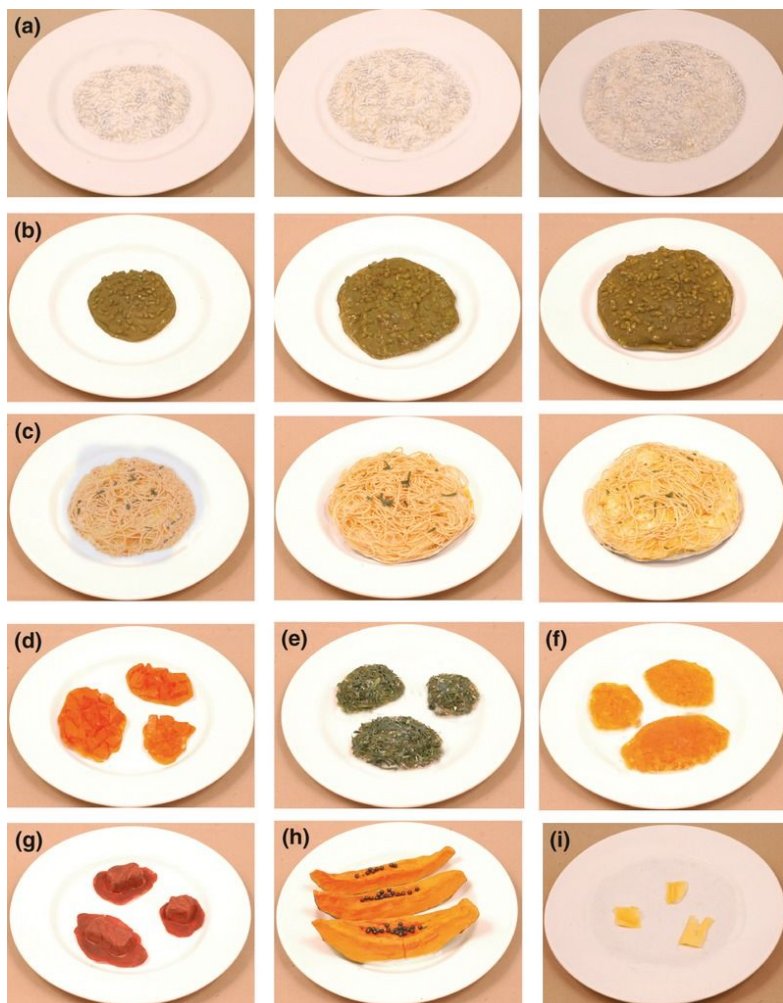


Photo credit: Stella Nyamsangia; photo was taken in Tanzania

Photographs of food models depicting different portion sizes. (a) Rice, (b) green gram, (c) noodles, (d) carrots, (e) dark green leafy vegetables, (f) lentils, (g) fish, (h) papaya, (i) butter.

Source: Lanerolle, 2013

Food frequency questionnaire

- A food frequency questionnaire (FFQ) consists of a finite list of foods and beverages with response categories to indicate usual frequency of consumption over a specified time period.
- FFQ rely on a longer recall period to capture foods that are not consumed every day but are still part of an individual's typical diet.
- FFQs can be self-administered or conducted by an interviewer, for example, when literacy is low.

Dietary diversity

- Dietary diversity can be measured at either the household or individual level.
- Household dietary diversity has been shown to be associated with caloric and protein adequacy and household income.
- Individual dietary diversity indices, specifically the **Minimum Dietary Diversity for Women** (MDD-W) and the **Minimum Dietary Diversity** (MDD) for children 6-23 months, have been shown to be a rough proxy for diet quality and nutrient adequacy.

Approaches to measure dietary diversity

Dietary Diversity Score	Data collection level	Number of food groups	Recall period	Purpose	Universal cut-off?	Promoted by
Household level measures						
Household dietary diversity score (HDDS)	Household	12	24 hours	Household dietary diversity and proxy for household food access and socioeconomic status	No	FAO
Food consumption score (FCS)	Household	8	7 days	Measures "usual" household consumption; Standardized food group weights are used to construct index.	Yes	WFP
Individual level measures						
Minimum Acceptable Diet (MAD)	Infant/child (6-23 months)	8 (from MDD)	24 hours	Measures both minimum dietary diversity and minimum meal frequency.	Yes	WHO
Minimum Dietary Diversity (MDD)	Infant/child (6-23 months)	8	24 hours	Measures infant and child dietary quality and adoption of complementary feeding practices	Yes	WHO
Minimum Dietary Diversity for Women (MDD-W)	(Individual) Women 15-49*	10	24 hours	Dichotomous indicator that measures the dietary diversity of an individual woman; associated with nutrient adequacy in many contexts and can be used as a proxy for overall diet quality	Yes	WHO, USAID

*Women of reproductive age

Source: <https://index.nutrition.tufts.edu/data4diets/data-source/dietary-diversity>

Vital health statistics as indirect nutritional assessment method

- An analysis of vital statistics such as mortality and morbidity data will identify groups at high risk and indicate the extent of that risk.
- Mortality in the under-5 age group is particularly related to malnutrition. Infant mortality rate, rate of low-birth-weight babies, and life expectancy can also be used as malnutrition indicators.
- These rates are influenced by nutritional status and are useful for understanding the nutritional status of a population. However, mortality data alone do not provide a satisfactory picture of the nutritional status of a population.
- Data on morbidity (e.g., hospital data or data from community health and morbidity surveys) particularly in relation to protein energy malnutrition, anemia, xerophthalmia, and other vitamin deficiencies, endemic goiter, diarrhea, measles, and parasitic infestations can be valuable in providing additional information about the nutritional status of the community.

Indirect dietary methods: Food balance sheets

- Food balance sheets (FBS) from the Food and Agriculture Organization of the United Nations (FAO) are used to assess the national **availability** of food for consumption.
- They are widely used in the food and agriculture sectors to monitor global food and dietary patterns.
- FAO FBS only provide data on annual per capita consumption of energy, protein, and fat (not micronutrients).
- FBS have also been used to set public health priorities, formulate policies, undertake inter-country comparisons, monitor trends over time, and estimate the likelihood of micronutrient deficiencies.

Household consumption and expenditure surveys (HCES)

- These surveys provide the total amount of food available for consumption in the household, generally excluding food eaten away from home.
- HCES can be used to track consumption patterns.
 - Consumption can vary among individuals within a household, especially by gender and age.
- They can also be used to monitor food security globally.
- Information on food consumption at the household level allows the derivation of variability parameters, such as the coefficient of variation of food consumption, which are used to estimate undernourishment in a population.

Takeaway messages

1. Nutritional assessment can identify individuals and populations at risk of malnutrition (over- and undernutrition).
2. Different methods of nutritional assessment (anthropometric, biochemical, clinical, and dietary assessment) can identify poor growth in children and risk of non-communicable diseases in adults.
3. Overall, nutritional assessment is essential for:
 - Identification of nutrient deficiencies and undernutrition in adults and children
 - Development of public health nutrition interventions, programs, and policies for nutrition-related disease prevention
 - Monitoring the efficacy and effectiveness of nutrition interventions, programs, and policies

Thank you!

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